ESP-Drone Setup Guide

From a fresh Windows PC to a flashed drone streaming live data to cfclient

This guide sets up the **Robocode frozen version** of ESP-Drone. Every part is pinned to an exact, tested version, so it works the same today and next year. Follow the parts in order; Parts A, B and D are done **once per PC**.

Part	What you do	Time
A	Install the tools: Git and ESP-IDF 5.0.7	25-40 min
B	Download the Robocode ESP-Drone code	2 min
C	Build the firmware and flash it to the drone	10 min
D	Install cfclient (PC app to view data and fly) - one double-click, also sets up Python	5-10 min
E	Connect to the drone and check live data	5 min
F	Troubleshooting	as needed

What you need

Item	Notes
Windows 10 or 11 PC	With internet for Parts A, B and D. Admin rights to install software.
ESP-Drone (ESP32-S2 or ESP32-S3)	The Robocode drones are ESP32-S2. Battery charged.
USB cable	Must be a **data** cable (some phone-charging cables are power-only).
Access to the code	A GitHub account with access to **RobocodeUK/esp-drone-idf5**, or a copy of the folder from your teacher.
Gamepad (optional)	For flying from cfclient, e.g. a PS3/PS4/Xbox USB controller.

> **IMPORTANT** Use exactly **ESP-IDF 5.0.7**: newer versions will not build this code. cfclient needs **Python 3.11**; the cfclient installer in Part D sets this up for you, so you do not need to install Python yourself. Do not download ESP-Drone from Espressif's GitHub; use the Robocode copy.

Part A - Install the tools (once per PC)

1 Install Git for Windows

Go to **git-scm.com/download/win**, download the 64-bit installer and run it. Click **Next** on every page to accept the defaults, then **Install**.

> **CHECK** Open **Command Prompt** (Start menu, type *cmd*) and run `git --version`. It should print a version number.

2 Install ESP-IDF 5.0.7 (the ESP32 build tools)

Go to **dl.espressif.com/dl/esp-idf/** and download the **ESP-IDF v5.0.7 Offline Installer** (file name similar to `esp-idf-tools-setup-offline-5.0.7.exe`, about 1 GB). Run it and:

- Keep the default install folder `C:\Espressif`.
- If it offers to **apply fixes** (e.g. long path support), accept.
- On the components page, leave the **drivers** ticked (Silicon Labs CP210x, FTDI, Espressif USB). These let Windows talk to the drone over USB.
- Leave the target chips ticked (at least ESP32-S2 and ESP32-S3).

Installation takes 15-30 minutes. At the end it adds Start menu shortcuts called **ESP-IDF 5.0 CMD** and **ESP-IDF 5.0 PowerShell**.

> **CHECK** Open **ESP-IDF 5.0 CMD** from the Start menu and run `idf.py --version`. It should print **ESP-IDF v5.0.7**.

> **IMPORTANT** Always build and flash from the **ESP-IDF 5.0 CMD** shortcut. A normal Command Prompt or PowerShell window does not have the tools set up and will show errors.

Part B - Download the code (once per PC)

1 Clone the code into C:\esp\esp-drone

Open **Command Prompt** and run these two commands:

```
mkdir C:\esp
git clone https://github.com/RobocodeUK/esp-drone-idf5.git C:\esp\esp-drone
```

The first time, a GitHub sign-in window may appear. Sign in with the account that has access to RobocodeUK.

> **NOTE** No GitHub access? Ask your teacher for a copy of the **esp-drone** folder and put it at `C:\esp\esp-drone`. Keep this exact short path: long paths or paths with spaces cause build errors on Windows.

Part C - Build and flash the firmware

1 **Plug in the drone and find its COM port**

Connect the drone to the PC with the USB data cable. Open **Device Manager** (right-click Start) and expand **Ports (COM & LPT)**. Look for **Silicon Labs CP210x USB to UART Bridge (COM3)**. The number (here **COM3**) is your port.

> **NOTE** If no new port appears: try another USB cable (it may be power-only) or another USB socket. Unplug and replug the drone and watch which entry appears.

2 **Open ESP-IDF 5.0 CMD and go to the code folder**

```
cd C:\esp\esp-drone
```

3 **Check which chip the drone has (ESP32-S2 or ESP32-S3)**

Replace COM3 with your port:

```
python -m esptool -p COM3 chip_id
```

Look for the line `Chip is ESP32-S2` or `Chip is ESP32-S3`.

4 **Select the chip type**

For ESP32-S2 (the Robocode drones):

```
idf.py set-target esp32s2
```

For ESP32-S3 use `idf.py set-target esp32s3` instead. You only need to do this again if you switch to a drone with the other chip.

5 **Build, flash and watch the drone start up**

```
idf.py -p COM3 flash monitor
```

The first build takes 5-10 minutes; later builds are much faster. When flashing finishes, the drone restarts and its log scrolls in the window.

> **CHECK** In the log you should see `MPU6050 I2C connection [OK]` and, after the drone has been kept still for a few seconds, `STAB: Ready to fly.` Lines saying VL53L1X, Z-down or PMW3901 **[FAIL]** are normal: those are optional add-on sensors that are not fitted.

Press **Ctrl +]** to leave the monitor. The drone is now flashed.

Part D - Install cfclient (once per PC)

1 Run the installer

cfclient is the PC app for viewing the drone's live sensor data and flying it with a gamepad. In File Explorer open `C:\esp\esp-drone\tools\cfclient` and double-click **install_cfclient.bat**. It works through four stages:

• **[1/4]** Checks for Python 3.11. If this PC does not have it, it downloads **Python 3.11.9** from python.org and installs it for your Windows user (no admin needed, and any other Python versions are left alone).
• **[2/4]** Checks Git is installed.
• **[3/4]** Installs cfclient with the exact tested versions of everything.
• **[4/4]** Adds a **cfclient (ESP-Drone)** shortcut to the desktop.

It takes 5-10 minutes. Leave the window open until it finishes.

> **CHECK** The window ends with **Done** and the **cfclient (ESP-Drone)** shortcut is on the desktop. Press any key to close the window.

> **NOTE** If it says *Git not found*, install Git (Part A step 1) and run it again. It is safe to run the installer again at any time.

Part E - Connect to the drone

> **SAFETY** For first tests, **remove the propellers** or hold the drone firmly. When a gamepad is connected, moving the thrust stick spins the motors.

1 Power on the drone and keep it still

Connect the battery (USB is not needed for flying). Put the drone on a flat surface and do not touch it for about 10 seconds while it calibrates its sensors.

2 Connect the PC to the drone's Wi-Fi

In the Windows Wi-Fi list, choose **ESP-DRONE_XXXXXXXXXXXX** (each drone has its own number) and enter the password **12345678**. Windows will say *No internet*; that is normal.

> **IMPORTANT** Windows may switch back to your normal Wi-Fi after about 40 seconds because the drone has no internet, which cuts off cfclient. To stop this, go to **Settings > Network & internet > Wi-Fi > Manage known networks**, click your normal network and untick **Connect automatically when in range**. Turn it back on when you finish.

3 Open cfclient and connect

If you are using a gamepad, plug it in **before** opening cfclient. Start **cfclient (ESP-Drone)** from the desktop. Click **Scan**, choose `udp://192.168.43.42:2390` in the list next to it, then click **Connect**.

> **CHECK** On the **Flight Control** tab, roll, pitch and yaw update live. Tilt the drone gently by hand and the numbers and horizon should follow it, and keep following it for minutes. The drone's green LED blinks quickly while connected.

4 Set up a gamepad (optional)

Menu **Input device**: select your controller, then choose a mapping (for a PS3/PS4 pad, **PS3_Mode_1** is a good start). Use **Input device > Configure device mapping** to change it. Keep the thrust stick fully down when connecting.

5 Finish

Click **Disconnect** in cfclient, unplug the drone battery, and reconnect the PC to your normal Wi-Fi.

Part F - Troubleshooting

Problem	Fix
`'idf.py' is not recognized`, or errors like *tool xtensa... has no installed versions* / *python_env ... doesn't exist*	You are in a normal Command Prompt. Open **ESP-IDF 5.0 CMD** from the Start menu and try again.
No COM port appears for the drone	Use a different (data) USB cable or USB socket. If it still does not appear, re-run the ESP-IDF installer and make sure the **drivers** are ticked.
Flashing fails: *Failed to connect* or *Access is denied*	Close anything else using the port (another monitor window, Arduino IDE). Check the COM number. If it still fails, hold the **BOOT** button on the drone board while the flash starts.
Drone log shows `MPU6050 I2C connection [FAIL]`. cfclient connects but the data never moves	The motion sensor has locked up (common after a crash or on a low battery). Unplug **both** the USB cable and the battery, wait 10 seconds, then power on again. Charge the battery.
cfclient data freezes after about 40 seconds	Windows switched back to your normal Wi-Fi. See the **IMPORTANT** box in Part E step 2.
cfclient data freezes after only a few seconds	You have an old cfclient install. Re-run **install_cfclient.bat** (Part D).
cfclient: *Socket is already connected* or will not reconnect	Close cfclient completely and open it again.
Scan / Connect does nothing	Check the PC is connected to the drone's Wi-Fi (not your normal network) and the drone is powered.
install_cfclient.bat: *Git not found*	Install Git for Windows (Part A step 1), then run it again.
install_cfclient.bat fails at step [1/4] (Python download or install)	Check the internet connection and run it again. If it still fails, install **Python 3.11.9** yourself from **python.org/downloads/release/python-3119/** (Windows installer 64-bit) and run the .bat again; it will find it.
Build errors mentioning long paths or missing files	Make sure the code is at exactly `C:\esp\esp-drone` (no spaces).

For teachers: keeping the setup frozen

Everything is pinned so nothing changes unless Robocode changes it:

Part	Pinned to
Firmware	**RobocodeUK/esp-drone-idf5** (based on Espressif esp-drone commit db0f6562, the last version for ESP-IDF 5.0). esp-now 2.5.2 is included in the code, so nothing is downloaded at build time.
cfclient	**RobocodeUK/crazyflie-clients-python**, commit 76d5093
cflib (drone link)	**RobocodeUK/crazyflie-lib-python**, commit f2baeab (adds the Wi-Fi keep-alive fix)
Python libraries	Exact versions in `tools\cfclient\requirements-cfclient.txt`
Tools	ESP-IDF 5.0.7, Python 3.11.9

• Do not update to Espressif's latest esp-drone: it needs ESP-IDF 5.5 or 6.1.
• To get Robocode updates later, run `git pull` inside `C:\esp\esp-drone`, then re-run **install_cfclient.bat**.
• More details are in `ROBOCODE_README.md` in the code folder.